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S. No.	Practical	Date of Execution	Date of Checking	Signature
DATA VISUALIZATION Lab.				
SECTION-A				
01.	Create a simple bar chart using student marks or sales data and explain what it shows.	23 rd January, 2026	23 rd January, 2026	
02.	Draw a line chart to represent time-based data (e.g., monthly attendance or sales).	23 rd January, 2026	23 rd January, 2026	
03.	Perform data representation vs data presentation for the same dataset using at least two different chart types.	30 th January, 2026	30 th January, 2026	
04.	Identify the purpose of data visualization for a given dataset and use a donut chart.	30 th January, 2026	30 th January, 2026	
05.	Explain the seven stages of data visualization using a simple example.	06 th February, 2026		
06.	Create one basic visualization using any data visualization tool (Excel / Google Sheets / Tableau / Power BI).	06 th February, 2026		
SECTION-B				
07.	Create an area chart for time-series data such as yearly profit or rainfall data.			
08.	Draw a scatter plot to show the relationship between two variables (e.g., study hours vs marks).			
09.	Create a pivot table and a pivot chart using a simple dataset.			
10.	Design a tree map to represent category-wise data (e.g., department-wise student count).			
11.	Draw a simple node-link diagram to show relationships (e.g., friends in a class).			
12.	Visualize the correlation between different numeric variables in a dataset using a color-coded matrix (heatmap).			
SECTION-C				
13.	Create a time-series visualization with multiple variables and analyze the observed patterns.			
14.	Perform text visualization by counting word frequency from a paragraph and plotting it.			
15.	Create a simple multivariate chart using more than two variables (e.g., marks in three subjects).			
16.	Study a small case example where visualization helps in understanding data easily.			
17.	Create a live-like data visualization using continuously updated values (manual update is allowed).			
18.	Write basic rules or best practices for creating good visualizations with examples.			

SECTION-D

19.	Create a basic chart to show open and closed ports from given data.			
20.	Create a firewall log visualization to identify suspicious traffic patterns.			
21.	Design an intrusion detection log visualization and highlight possible attacks.			
22.	Write a program to implement sentiment analysis on the live tweet data using Python libraries.			
23.	Write a program to create interactive visualizations to monitor device activity for Pre-processed simulated IoT data streams in real-time.			
24.	Design a secured data visualization system architecture and explain the security measures used.			

Beyond Syllabus: Practical Assignments Beyond Syllabus

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Ms. Chetna Thakur
Assistant Professor